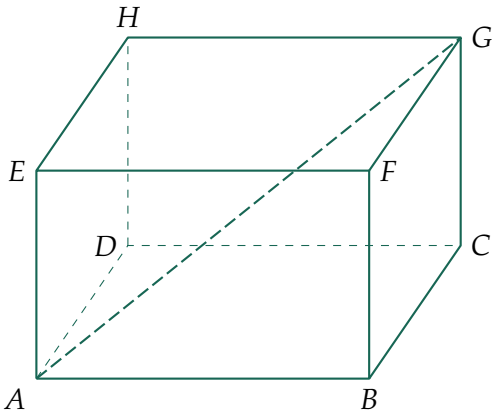
Research activity

Three midpoints in a cube — 40 min.

Consider the cube $ABCDEFGH$ shown below. Let $\vec{\imath} = \overrightarrow{AB}$, $\vec{\jmath} = \overrightarrow{AD}$ and $\vec{k} = \overrightarrow{AE}$. The points I, J and K are the midpoints of $[AB]$, $[AD]$ and $[AE]$ respectively. Let O be the centre of the cube.



$[AG]$ is a space diagonal, drawn dotted in green.
The hidden edges are thin dashed lines; D and F are not on (AG).

1. Express $\overrightarrow{IJ}$ and $\overrightarrow{IK}$ in the basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$.
2. Prove that the points I, J and K are not collinear.
3. Determine whether O belongs to the plane (IJK). Justify the answer by investigating whether $\overrightarrow{IO}$ is a linear combination of $\overrightarrow{IJ}$ and $\overrightarrow{IK}$.
4. Why does simply reading the figure not allow one to prove that O does or does not belong to the plane (IJK)?